A Novel Technique for Estimation and Control of Stator Flux of a Salient-Pole PMSM in DTC Method Based on MTPF

Jawad Faiz, *Senior Member, IEEE*, and S. Hossein Mohseni-Zonoozi

***Abstract*—A permanent-magnet synchronous machine (PMSM) can be controlled using the direct torque control (DTC) technique in three different ways, i.e., by controlling flux, reactive torque and rotor d-axis current. Frequently, the DTC technique controls the speed of the motor by controlling stator flux with the aim of obtaining an optimal torque. A varying flux, proportional to the torque, may be used instead of a fixed flux, resulting in a maximum torque per ampere or maximum torque per flux (MTPF).**

In this paper, a reference-flux-generating method is followed to achieve the MTPF. An approximate equation is then derived using numerical techniques in order to obtain the reference flux from the torque. This equation is then applied to the DTC control system in order to obtain the reference flux. The control scheme has been verified by simulation and tests on a salient-pole permanent-magnet synchronous motor.

***Index Terms*—Analytical technique, direct torque control, permanent-magnet synchronous motor (PMSM), salient-pole motor.**

NOMENCLATURE

d, q	Rotor direct and quadrature axis indexes.
λ_f	Rotor flux (PM).
L_d, L_q	d and q axes synchronous inductances.
i_d, i_q	d and q axes currents.
λ_d, λ_q	d and q axes flux linkages.
λ_s	Stator flux linkage.
V_d, V_q	d axis and q axis voltages.
i_s	Stator current.
T_e	Electromagnetic torque.
T_q	Reactive torque.
ω_r	Rotor speed.
p	Operator d/dt.
T_{cogg}	Cogging torque.
T_B	Friction torque.
B	Friction coefficient.
T_J	Moment torque.
J	Moment of inertia.

Manuscript received July 27, 2001; revised July 22, 2002. Abstract published on the Internet February 4, 2003. This work was supported by the University of Tabriz, Tabriz, Iran.

J. Faiz is with the Department of Electrical and Computer Engineering, University of Tehran, 14395 Tehran, Iran.

S. H. Mohseni-Zonoozi was with the Department of Electrical Engineering, University of Tabriz, 51664 Tabriz, Iran.

Digital Object Identifier 10.1109/TIE.2003.811591

I. INTRODUCTION

THE availability of high-speed digital signal processors (DSPs) has resulted in the wide application of the direct torque control (DTC) technique in induction motor drives. The basic principle of DTC involves selecting stator voltage vectors according to the differences between the reference and actual torques and stator flux linkage [1], [2]. The DTC technique possesses advantages such as less parameter dependency and fast torque response when compared to the torque control via pulsewidth modulation (PWM) current control.

There are few papers concerning the application of DTC to permanent-magnet synchronous motor (PMSM) drives. In one such paper, a quick torque response has been obtained by adjusting the rotating speed of the stator flux linkage as fast as possible [3]. The DTC technique has then been applied to an interior PMSM (IPMSM) drive [4]; and an important difference in switching table for IPMSM, compared to induction motors, has been found. A direct torque controller may be used in which a torque estimator for the feedback to the torque control loop is employed [5]. However, current controllers and continuous position feedback are also required in this method.

In this paper, a new analytical technique is introduced for generating the reference flux from the torque. It is shown how the maximum torque per ampere (MTPA) can be followed in the control process. Different criteria are considered for obtaining the reference flux of motors with a large magnetic time constant and motors with high speed. Contrary to previous publications, here, a salient-pole PMSM motor is simulated using the maximum torque per flux (MTPF), and the reference flux determined. The analytically calculated flux from the torque has a good approximation; therefore, the lookup table is not required.

This paper is organized as follows. In Section II, expressions are presented for the currents of the PMSM in the stator flux reference frame (xy). In Section III, the torque equations are derived for a salient-pole PMSM in terms of the stator flux-linkage and its angle with respect to the rotor flux linkage. The maximum load angle in a PMSM, at which the maximum load torque exists, is obtained in Section IV. In Section V, the flux reference frame for the MTPF is determined. Simulation results using the proposed method are presented in Section VI. Experimental results are then presented in Section VII, in order to confirm the simulated results. Conclusions are presented in Section VIII.

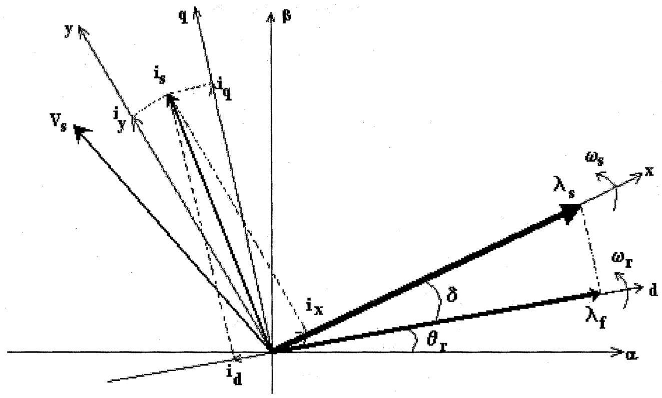

Fig. 1. Stator and rotor flux linkage in $\alpha\beta$, dq, and xy reference frames.

II. PMSM Equations

The stator flux linkage, voltage, and electromagnetic torque equations in the rotor flux reference frame for a PMSM can be expressed as follows:

$$\begin{aligned}\lambda_d =&L_d i_d + \lambda_f\\ \lambda_q =&L_q i_q\\ \lambda_s =&\sqrt{\left(\lambda_d^2+\lambda_q^2\right)}\end{aligned} \tag{1}$$

$$\begin{aligned}v_d =&R_s i_d + p\lambda_d - \omega_r\lambda_q\\ v_q =&R_s i_q + p\lambda_q + \omega_r\lambda_d\end{aligned} \tag{2}$$

$$\begin{aligned}T_e =&P(\lambda_d i_q - \lambda_q i_d)\\ T_q =&P(\lambda_d i_d + \lambda_q i_q).\end{aligned} \tag{3}$$

Considering the dynamics of the motor

$$\begin{aligned}T_e =&T_B + T_{\text{cogg}} + J\frac{d\omega_r}{dt}\\ =&B\omega_r + \sum_{i=1}^{4} T_{ri}\text{Sin}(6i\theta_r) + J\frac{d\omega_r}{dt}.\end{aligned} \tag{4}$$

Fig. 1 shows the stator flux-linkage vector and rotor flux-linkage vector in the rotor flux (dq), stator flux (xy), and stationary $(\alpha\beta)$ reference frames. δ is the angle between the stator and rotor flux linkage, which becomes the load angle if R_s is neglected. Referring to Fig. 1, the flux linkage and voltage in (1) and (2) can be transformed to the stator flux xy reference frame, as follows [3]:

$$\begin{bmatrix}F_x\\F_y\end{bmatrix} = \begin{bmatrix}\text{Cos}\delta & \text{Sin}\delta\\ -\text{Sin}\delta & \text{Cos}\delta\end{bmatrix}\begin{bmatrix}F_d\\F_q\end{bmatrix}. \tag{5}$$

Using (1), the current matrix is

$$\begin{bmatrix}i_d\\i_q\end{bmatrix} = \begin{bmatrix}\frac{1}{L_d} & 0\\ 0 & \frac{1}{L_q}\end{bmatrix}\begin{bmatrix}\lambda_d\\ \lambda_q\end{bmatrix} - \frac{\lambda_f}{L_d}\begin{bmatrix}1\\0\end{bmatrix}. \tag{6}$$

Therefore, components of the stator current in the stator flux reference frame will be as follows:

$$\begin{aligned}\begin{bmatrix}i_x\\i_y\end{bmatrix} =& \begin{bmatrix}\frac{\text{Cos}^2\delta}{L_d}+\frac{\text{Sin}^2\delta}{L_q} & \left(\frac{1}{L_q}-\frac{1}{L_d}\right)\text{Cos}\delta\text{Sin}\delta\\ \left(\frac{1}{L_q}-\frac{1}{L_d}\right)\text{Cos}\delta\text{Sin}\delta & \frac{\text{Cos}^2\delta}{L_q}+\frac{\text{Sin}^2\delta}{L_d}\end{bmatrix}\\ &\times\begin{bmatrix}\lambda_x\\ \lambda_y\end{bmatrix} - \frac{\lambda_f}{L_s}\begin{bmatrix}\text{Cos}\delta\\ -\text{Sin}\delta\end{bmatrix}.\end{aligned} \tag{7}$$

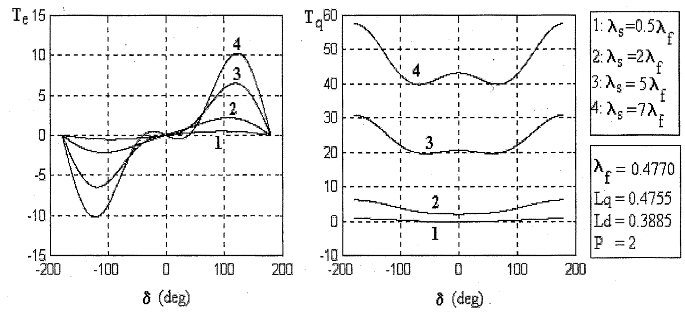

Fig. 2. Variation of T_e and T_q versus δ for different λ_s.

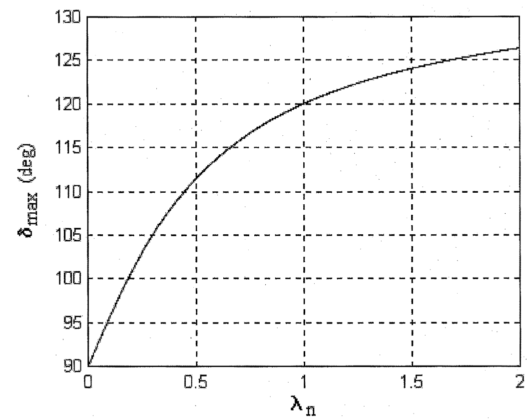

Fig. 3. Variation of $\delta_{\max}$ versus normalized flux λ_n.

III. Torque Equations in the Stator Flux Reference Frame

Referring to Fig. 1, the following equations can be obtained for a salient-pole PMSM [3]:

$$\text{Sin}\delta = \frac{\lambda_q}{|\lambda_s|}, \quad \text{Cos}\delta = \frac{\lambda_d}{|\lambda_s|} \tag{8}$$

$$\begin{aligned}T_e =&P\left(\lambda_d(i_x\text{Sin}\delta + i_y\text{Cos}\delta) - \lambda_q(i_x\text{Cos}\delta - i_y\text{Sin}\delta)\right)\\ =&P\left(i_x\frac{\lambda_d\lambda_q}{|\lambda_s|} + i_y\frac{\lambda_d^2}{|\lambda_s|} - i_x\frac{\lambda_d\lambda_q}{|\lambda_s|} + i_y\frac{\lambda_q^2}{|\lambda_s|}\right)\\ =&P\frac{\lambda_d^2+\lambda_q^2}{|\lambda_s|}i_y = P\,|\lambda_s|\ i_y.\end{aligned} \tag{9}$$

Similarly, the reactive torque T_q can be determined as follows:

$$T_q = P|\lambda_s|i_x. \tag{10}$$

Considering $\lambda_y = 0$ and $\lambda_x = |\lambda_s|$ and substituting (7) into (9) and (10) leads to

$$\begin{aligned}i_x =& \left(\frac{\text{Cos}^2\delta}{L_d} + \frac{\text{Sin}^2\delta}{L_q}\right)|\lambda_s| - \frac{\lambda_f}{L_d}\text{Cos}\delta\\ i_y =& \left(\frac{1}{L_q} - \frac{1}{L_d}\right)|\lambda_s|\text{Cos}\delta\text{Sin}\delta + \frac{\lambda_f}{L_d}\text{Sin}\delta\end{aligned} \tag{11}$$

$$\begin{aligned}T_e =&\frac{P|\lambda_s|}{2L_dL_q}(2\lambda_f L_q\text{Sin}\delta - (L_q - L_d)|\lambda_s|\text{Sin}2\delta)\\ T_q =&\frac{P|\lambda_s|}{2L_dL_q}\left[\left(L_q\text{Cos}^2\delta + L_d\text{Sin}^2\delta\right)|\lambda_s|\right.\\ &\left.- L_q\lambda_f\text{Cos}\delta\right].\end{aligned} \tag{12}$$

Variations of $T_e(\delta)$ and $T_q(\delta)$ for different λ_s are presented in Fig. 2.

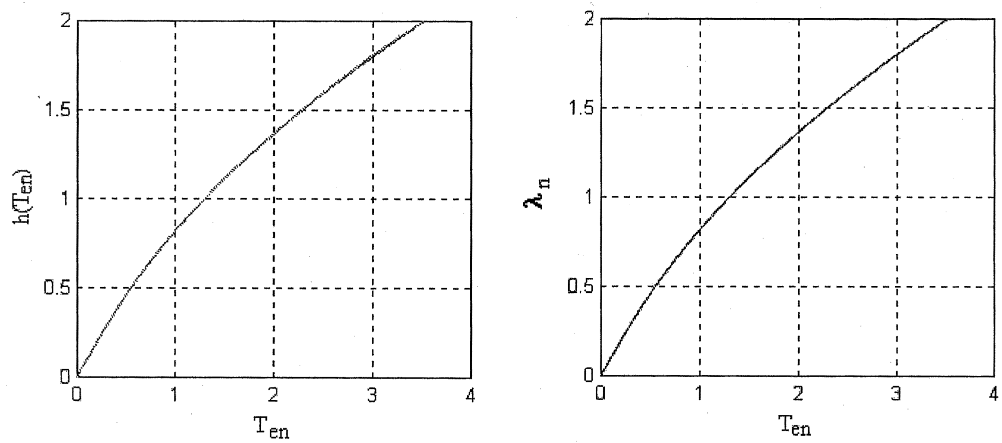

Fig. 4. Variation of λ_n and $h(T_{en})$ versus T_{en}.

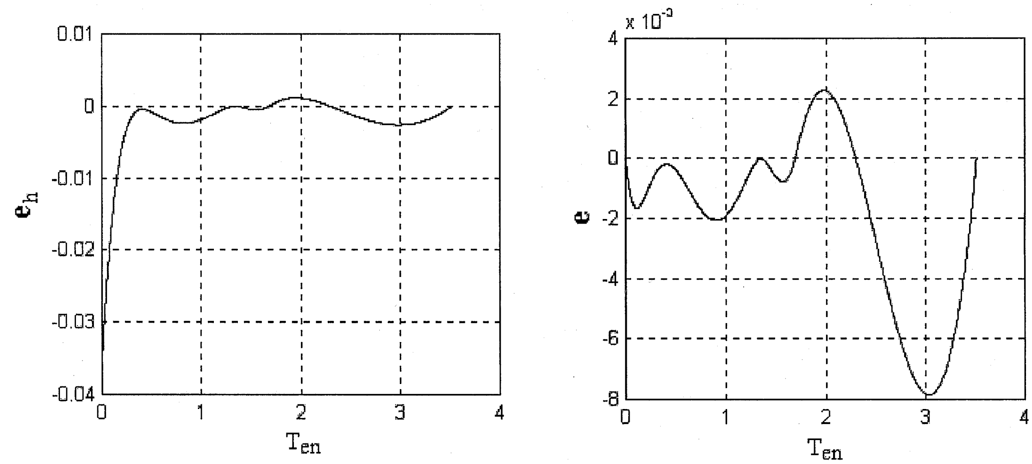

Fig. 5. Absolute error e and percentage error versus T_{en}.

TABLE I
CONSTANTS FOR (36)

T_{en0}	T_{en1}	T_{en2}	c_1	c_2	c_3	c_4	k
$3\sqrt{3}/a^2$	1.35	1.69	0.2622	0.5681	-0.1269	0.1454	$1/a\text{-}(c_1+c_2T_{en0}+c_3\|T_{en0}-T_{en1}\|)^{2.41}$

The condition for positive torque in the DTC technique is as follows [3]:

$$|\lambda_s| \le \frac{L_q}{L_q - L_d}\lambda_f = a\lambda_f \tag{13}$$

where $a = L_q/(L_q - L_d)$ and the maximum load torque under the fixed flux λ_s occurs at

$$\delta_{\max} = \mathrm{Cos}^{-1}\left(\frac{a\lambda_f}{4|\lambda_s|} - \frac{1}{2}\sqrt{\left(\frac{a\lambda_f}{2|\lambda_s|}\right)^2 + 2}\right). \tag{14}$$

IV. Maximum Load Angle Characteristic for a Salient-Pole PMSM

$|\lambda_s|$ is defined as follows:

$$|\lambda_s| = k\frac{L_q}{L_q - L_d}\lambda_f = k\,a\,\lambda_f. \tag{15}$$

Substituting (15) into (14) leads to

$$\delta_{\max} = \mathrm{Cos}^{-1}\left(\frac{1}{4k} - \frac{1}{2}\sqrt{\left(\frac{1}{2k}\right)^2 + 2}\right). \tag{16}$$

Equation (16) shows that $\delta_{\max}$ depends only on k, and not on parameters of the machine. Here, a new variable called normalized flux is introduced as follows:

$$\lambda_n = k = \frac{\lambda_s}{a\lambda_f} \tag{17}$$

and condition (13) will be as follows:

$$0 < \lambda_n \le 1 \tag{18}$$

and $\delta_{\max}$ will be as follows:

$$\delta_{\max} = \mathrm{Cos}^{-1}\left(\frac{1}{4\lambda_n}\left(1 - \sqrt{8\lambda_n^2 + 1}\right)\right) = f(\lambda_n). \tag{19}$$

If condition (18) is not taken into account, function $\delta_{\max}(\lambda_n)$ will be as shown in Fig. 3. In order to satisfy condition (18), a

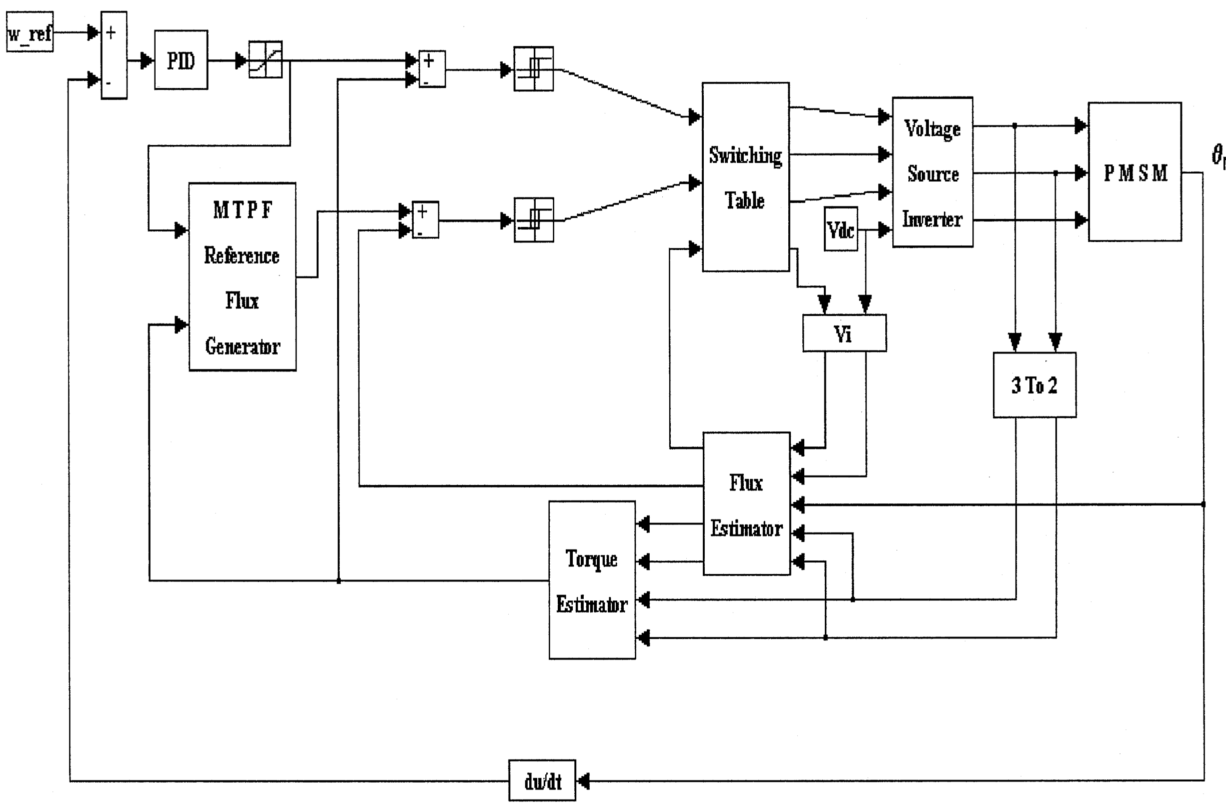

Fig. 6. Block diagram of the proposed DTC for speed control of a PMSM.

TABLE II
SPECIFICATIONS OF THE PROPOSED PMSMs

Motor		PMSM1	PMSM2	
Number of pole pairs	P	2	2	
Stator winding resistance	R_s	0.57	19.4	Ω
d-axis inductance	L_d	0.0087	0.3885	mH
q-axis inductance	L_q	0.0228	0.4755	mH
Magnetic flux-linkage	λ_f	0.108	0.447	Wb
Maximum phase current	I_{max}	13	1.6	A
Phase voltage	V_{ph}	50	240	V
Rotor speed	ω_r	110	157	rad/s

proper part of the curve within the limits given by $0 < \lambda_n \leq 1$, is considered. If λ_n tends to infinity, $\delta_{\max}$ tends to 135°. For λ_n equal to 1, $\delta_{\max}$ is 120°. As shown in Fig. 2, condition (18) must be satisfied in order to vary torque versus load angle at a constant flux. If flux can be varied proportional to torque, such that the flux is small enough over the low load angles and large enough over the high load angles, condition (18) will automatically be held. Thus, condition (18) cannot put a limit on the flux, but the peak current and saturation may be deterministic.

V. DETERMINATION OF REFERENCE FLUX FRAME FOR MTPF

One of the optimal methods in the DTC technique is selecting the stator flux from the motor or reference torque, such that the minimum current produces the maximum torque (MTPA). An alternative method is producing the minimum flux for developing maximum torque (MTPF). The MTPA and MTPF have been analytically applied to a rotor configuration, which does not possess saliency [6]. The implementation of PM synchronous mode for motors with PM rotors possessing saliency is quite different to that for motors without saliency. The MTPA for a salient-pole motor has been determined using a lookup table [3]. However, an analytical equation has not been given for a salient-pole PMSM in order to derive the flux from the torque.

In order to obtain the MTPA, the relationship between dq currents are given as follows [4]:

$$i_d = \frac{\lambda_f}{2(L_d - L_q)} - \sqrt{\frac{\lambda_f^2}{4(L_d - L_q)^2 + i_q^2}}. \tag{20}$$

In the present work, the relationship between λ_q and λ_d is obtained in order to achieve the MTPF. An analytical expression is then presented to derive flux from the torque.

1) *Relationship between λ_q and λ_d for MTPF*

Substituting (1) into (3) and eliminating the current leads to

$$\lambda_q = \frac{T_e}{[P(k_1\lambda_d + k_2)]} \tag{21}$$

where $k_1 = (1/L_q - 1/L_d)$ and $k_2 = \lambda_f/L_d$.

Since

$$\lambda_s^2 = \lambda_d^2 + \lambda_q^2 = \frac{\lambda_d^2 + T_e^2}{[P^2(k_1\lambda_d + k_2)^2]}. \tag{22}$$

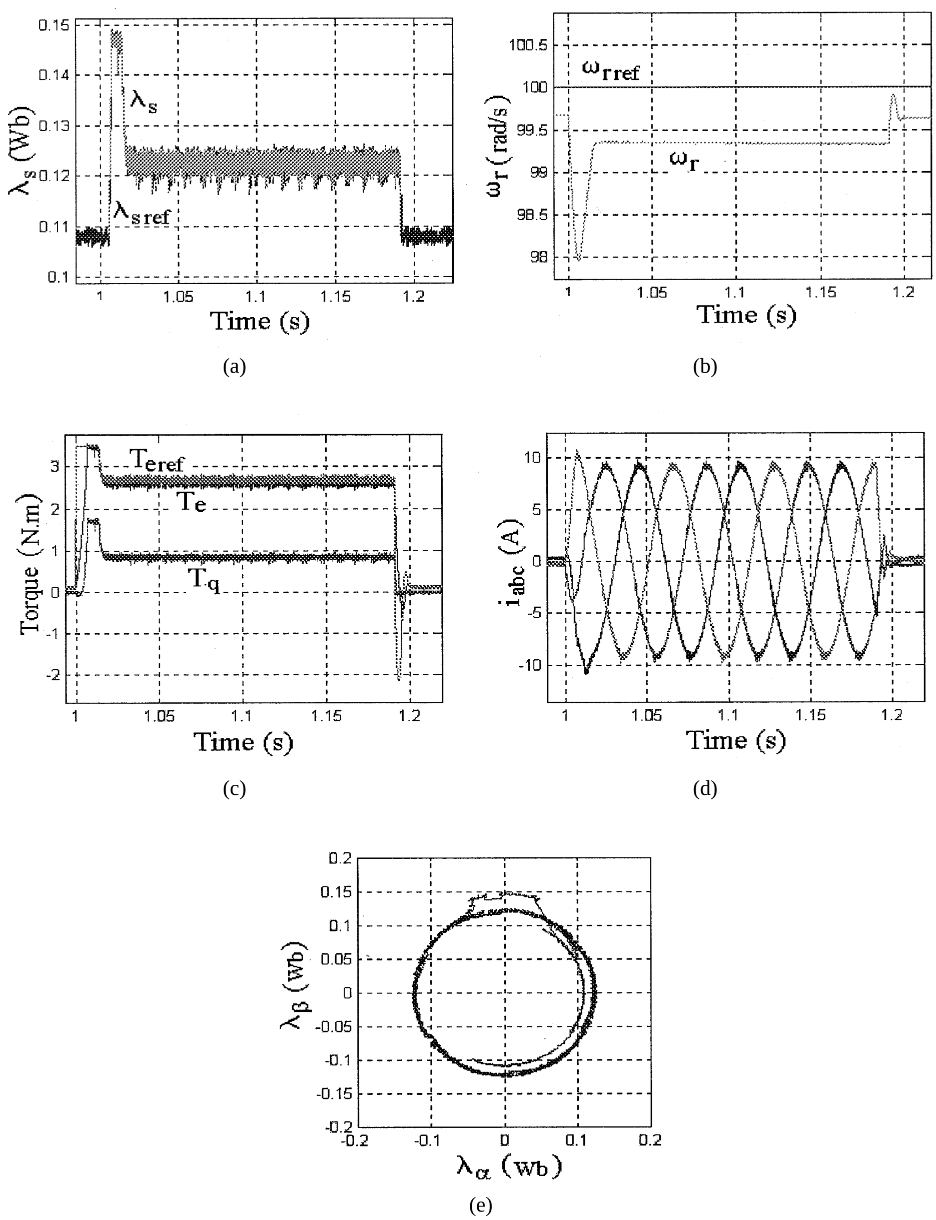

Fig. 7. Time variation of (a) stator flux, (b) speed, (c) torque, (d) three-phase currents and (e) locus of stator flux.

To achieve the MTPF, at a fixed torque, λ_s must be minimized. Thus, a relationship between λ_d and λ_q is found such that λ_s is minimized

$$\frac{d\lambda_s^2}{d\lambda_d} = 2\lambda_d + \frac{-2T_e^2 k_1}{[P^2(k_1\lambda_d + k_2)^3]} = 0. \tag{23}$$

Thus,

$$\frac{T_e^2}{P^2} = \frac{\lambda_d(k_1\lambda_d + k_2)^3}{k_1}. \tag{24}$$

On the other hand, (21) gives

$$\frac{T_e}{P} = \lambda_q(k_1\lambda_d + k_2). \tag{25}$$

Substituting (25) into (24) yields

$$\lambda_q^2 = \lambda_d^2 - a\lambda_f\lambda_d. \tag{26}$$

Since

$$\frac{\lambda_q^2 + a^2\lambda_f^2}{4} = \left(\frac{\lambda_d - a\lambda_f}{2}\right)^2. \tag{27}$$

Thus,

$$\lambda_d = \frac{a\lambda_f}{2 - \sqrt{\lambda_q^2}} + \frac{a^2\lambda_f^2}{4}. \tag{28}$$

Equation (28) relates λ_d and λ_q for MTPF and it is similar to (20), but not the same.

2) *Analytical equation of λ_s versus T_e for MTPF*

It is not a straightforward process to obtain an analytical equation for λ_s, therefore, a particular technique is followed in order to obtain an approximate analytical equation.

The angle at which the maximum torque for a fixed flux $|\lambda_s|$ is developed has already been defined by (14).

Moreover, (21) defined the variation of $\delta_{\max}$ with λ_n. Using (12), $T_{e\,\max}$ can be expressed as follows:

$$T_{e\,\max} = \frac{P\,|\lambda_s|}{2L_d L_q} \times \left(2\lambda_f L_q \mathrm{Sin}\delta_{\max} - (L_q - L_d)\,|\lambda_s|\,\mathrm{Sin}2\delta_{\max}\right). \quad (29)$$

Since $\delta_{\max}$ depends only on λ_n, then $\delta_{\max} = f(\lambda_n)$ and, thus, (29) can be rewritten as follows:

$$T_{e\,\max} = \left[\frac{Pa\lambda_f^2}{L_d}\right] \lambda_n \mathrm{Sin}(f(\lambda_n))[1 - \lambda_n \mathrm{Cos}(f(\lambda_n))]. \quad (30)$$

The normalized torque is defined as follows:

$$T_{en} = \frac{T_e L_d}{Pa\lambda_f^2} \quad (31)$$

$$T_{en} = \lambda_n \mathrm{Sin}\left(f(\lambda_n)\right)\left[1 - \lambda_n \mathrm{Cos}\left(f(\lambda_n)\right)\right] = g(\lambda_n). \quad (32)$$

As can be seen from the expressions above, the normalized flux and torque do not depend on the parameters of the machine.

However, determination of the inverse equation $\lambda_n = h(T_{en})$ is difficult. The variation of T_{en} with λ_n can be obtained by varying λ_n from 0 to 2. A curve-fitting technique can then be used to derive the equation $h(T_{en})$ as shown by (33), at the bottom of the page.

Fig. 4 shows the variation of $\lambda_n(T_{en})$ and $h(T_{en})$. It can be seen that $h(T_{en})$ is a good approximation for λ_n. The error between the actual and approximation values is

$$e = \lambda_n - h\left(T_{en}\right) \quad (34)$$

and percentage of the error is

$$e_{h(T_{en})} = \frac{\lambda_n(T_{en}) - h(T_n)}{\lambda_n(T_{en})} \times 100. \quad (35)$$

Fig. 5 exhibits the variations of absolute error e and its percentage e_h with T_{en}. A curve-fitting technique is used and equation $h(T_{en})$ is derived as described in Section VI.

As (33) confirms over a range of T_{en}, the value of λ_n becomes less than $1/a$. This means that over the specified range of the torque, $\lambda_s < \lambda_f$ and (33) must be modified. Since the stator flux is equal to the rotor flux at zero torque, λ_n must be taken as $1/a$ over the above range. Furthermore, when the torque approaches zero, the motor current has to be minimum. This occurs only at zero torque, when λ_s is equal to λ_f. Thus, the error for λ_n below the value of $1/a$ is removed. There will be errors around 0.01%. Referring to Fig. 4, for the region $0 < \lambda_n \leq 1$, $1/a < T_{en} < 1.5$, and as shown in Fig. 5, for this range of T_{en}, $-2 \times 10^{-3} < e < 0$ and $-0.005\% < e_h < 0\%$.

Thus, (33) is converted as shown in (36), at the bottom of the page, where k gives a continuity at point $T_{en} = T_{en0}$. The obtained values of c_1–c_4 and k have been listed in Table I. Although the method is approximate, it is an excellent technique for determining the reference flux and satisfying the MTPF.

The algorithm of the present method is summarized as follows.

1. T_{en} is obtained from T_e using $T_{en} = (|T_e|L_d)/(Pa\lambda_f^2)$.
2. λ_n is determined from $\lambda_n = h(T_{en})$.
3. λ_s is obtained from $\lambda_s = a\lambda_f\lambda_n$.

To determine the reference λ_s, three choices exist for obtaining T_e. These include the following:

1) output reference torque, $T_{e\text{-ref}}$ from the speed controller, where condition (18) must be satisfied;
2) instantaneous torque of motor $T_{e\text{-est}}$, which may lead to instability;
3) geometrical mean of (1) and (2), $\sqrt{T_{e\text{-ref}}\,T_{e\text{-est}}}$.

In order to overcome to the problems of 1) and 2), option 3) has been used in the present work. The block diagram of the proposed DTC for speed control of a PMSM is shown in Fig. 6. The three-phase variables are transformed to the stationary two-phase variables. The reference torque is the output of the speed controller, which is generated based on the speed errors. The output of the controller is limited to $\pm T_{\max}$ which is determined by the maximum current of the motor.

VI. Simulation Results

Two PMSMs are used for the simulation. Their specifications are given in Table II [3], [4].

A. Simulations for PMSM1

Initially, the motor operates at steady-state speed equal to -100 rad/s (CCW). Then, at $t = 0.4$ s, the speed changes from -100 rad/s to 100 rad/s (CW). Fig. 7 shows the time variations of the stator flux, speed, electromagnetic torque, three-phase currents, and the locus of the stator flux. Before applying the

$$h\left(T_{en}\right) = \begin{cases} c_1 + c_2\,|T_{en}| + c_3\left(||T_{en}| - T_{en1}|\right)^{2.41} & T_{en} \leq T_{en2} \\ c_1 + c_2\,|T_{en}| + c_3\left(||T_{en}| - T_{en1}|\right)^{2.41} + c_4\left(|T_{en}| - T_{en2}\right)^{2.22} & T_{en}\rangle T_{en2} \end{cases} \quad (33)$$

$$h(T_{en}) = \begin{cases} \frac{1}{a} & T_{en} \leq \frac{3\sqrt{3}}{a^2} = T_{en0} \\ c_1 + c_2|T_{en}| + c_3\left(||T_{en}| - T_{en1}|\right)^{2.41} + kT_{en} \leq T_{en2} & \\ c_1 + c_2|T_{en}| + c_3\left(||T_{en}| - T_{en1}|\right)^{2.41} + c_4\left(|T_{en}| - T_{en2}\right)^{2.22} + kT_{en}\rangle T_{en2} & \end{cases} \quad (36)$$

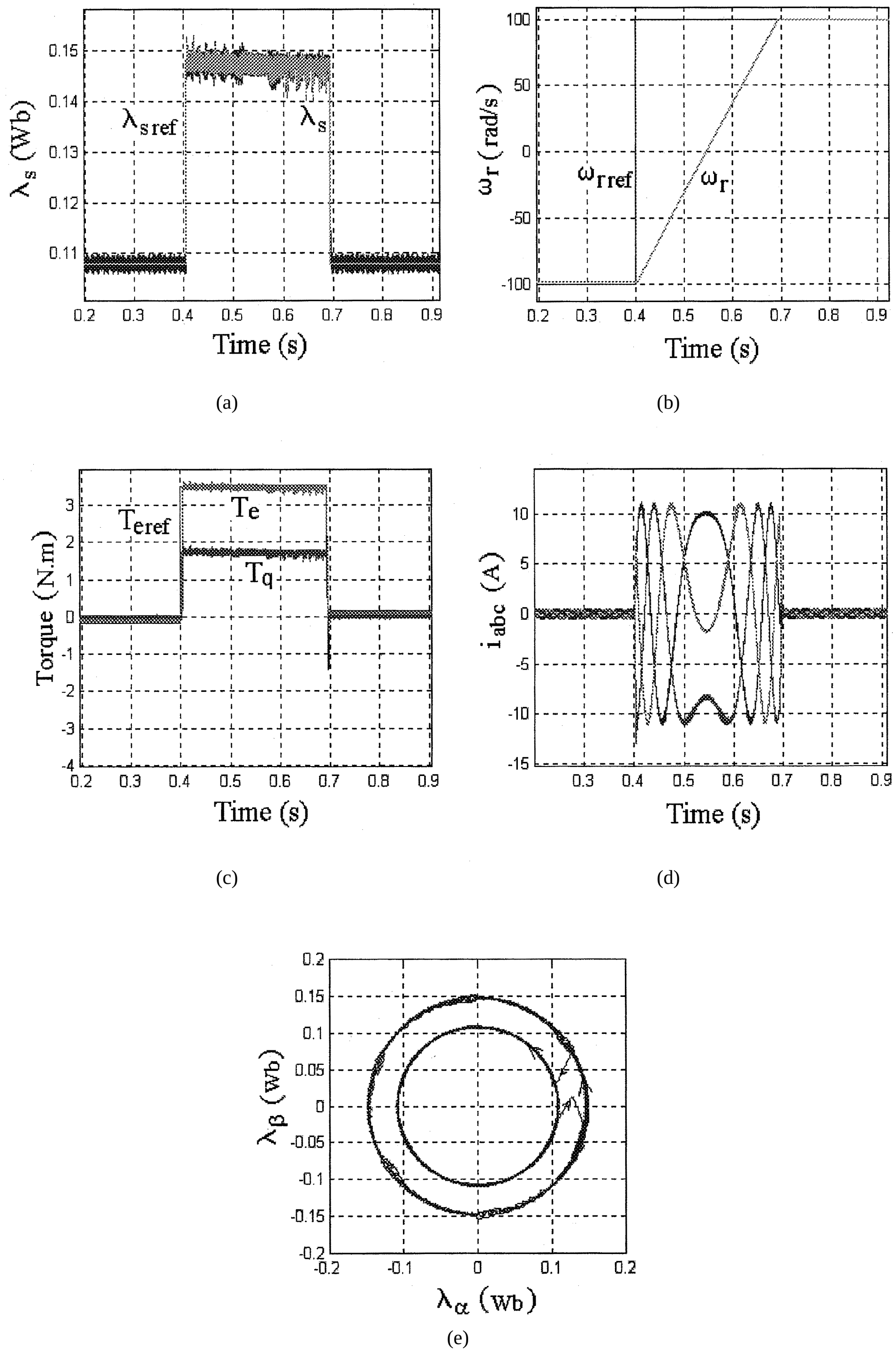

Fig. 8. Time variation of (a) stator flux, (b) speed, (c) torque, (d) three-phase currents, and (e) locus of stator flux when 2-N·m load applied.

step speed at $t = 0.4$ s, the motor was rotating with no load at speed -100 rad/s, thus the torque is almost equal to zero. Considering the MTPF, the steady-state stator flux at no load was λ_f, and after applying the speed command it approaches the reference value.

Fig. 8 shows the time variations of flux, speed, torque, three-phase currents, and locus of the stator flux when applying a load torque of 2 N·m to the motor. It can be seen that the motor responds quickly with a steady-state error of 0.5%, despite the applied load being larger than $T_{\max}/2$. Since the reference speed is 100 rad/s and the steady-state speed before applying the load is 99.7 rad/s, the speed after applying the load reaches 99.3 rad/s or 0.4% error.

B. Simulations for PMSM2

This motor has been tested using an MTPA technique in [3], [8]. The same motor is simulated using MTPF and the results are compared with [3] and [8]. In both techniques, the motor approaches its steady-state speed equal to 1200 r/min and then the load is increased. Fig. 9 presents the time variation of

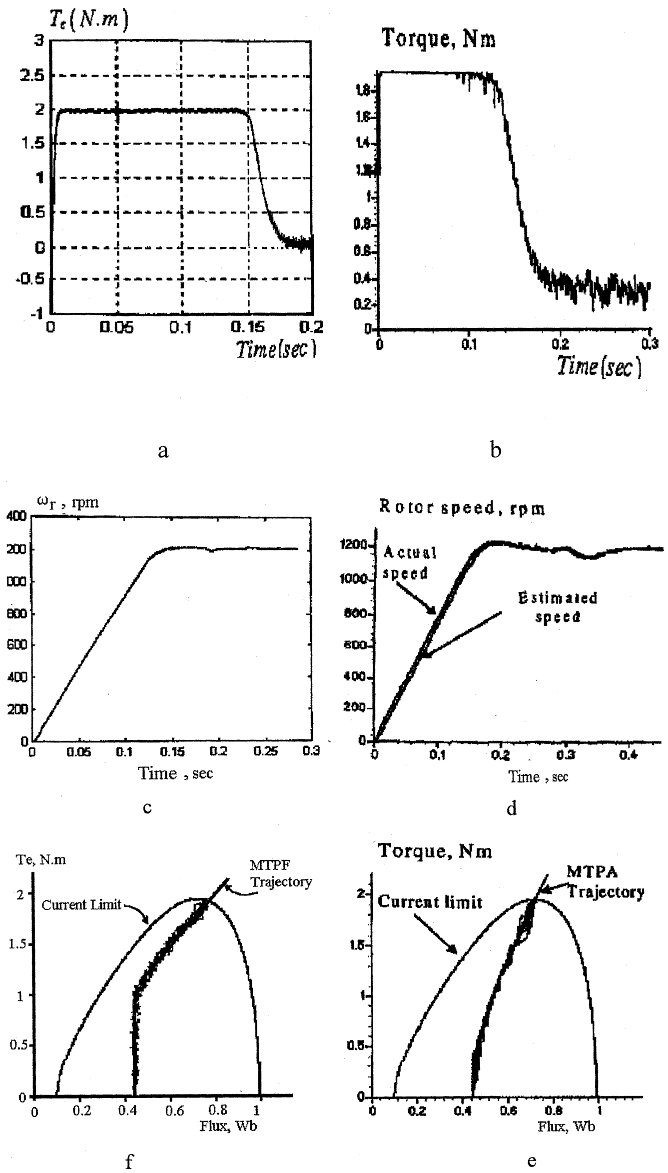

Fig. 9. Time variation of (a), (b) torque; (c), (d) speed; and (e), (f) trajectories of MTPA and MTPF.

torque, speed, and the MTPA and MTPF trajectories for both techniques. As shown in Fig. 9(e), $\lambda_n = 1/a$ (or $\lambda_s = \lambda_f$), when $T_{en} < T_{en0}$.

VII. Experimental Results

The proposed DTC drive of Fig. 6 was implemented for a PMSM with parameters summarized in Table III. The test setup is shown in Fig. 10.

A Pentium 200 MMX is used for control purposes. In considering the required sampling time with the available A/D converter and the computation and estimation times, a maximum switching frequency of 13 kHz is used. The measured or calculated values have been entered into a CRT by D/A converter. The sampling of currents has been achieved by means of a Hall-effect sensor.

Fig. 11 shows a typical line voltage and phase current waveforms when the steady-state speed of the motor approaches 800 r/min then it rises to 1600 r/min. Fig. 12 presents the simulation results with the real dc supply and rotor flux. Fig. 13 presents the corresponding experimental results. Comparison of Figs. 12 and 13 shows good agreement between the simulated and test

TABLE III
Parameters of Tested PMSM

Number of pole pairs	P	6	
Stator winding resistance	R_s	1.6	Ω
d-axis inductance	L_d	0.0187	mH
q-axis inductance	L_q	0.0489	mH
Magnetic flux-linkage	λ_f	0.140	Wb
Maximum phase current	I_{max}	15	A
Phase voltage	V_{ph}	100	V
Rotor speed	ω_r	1600	rpm

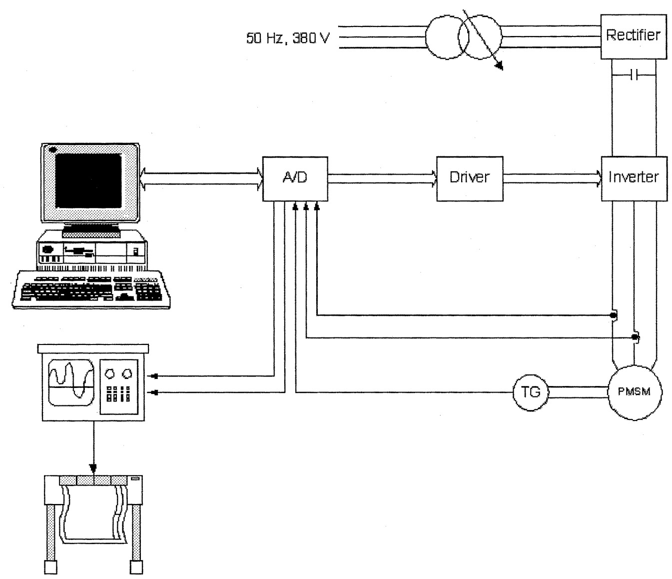

Fig. 10. Test setup.

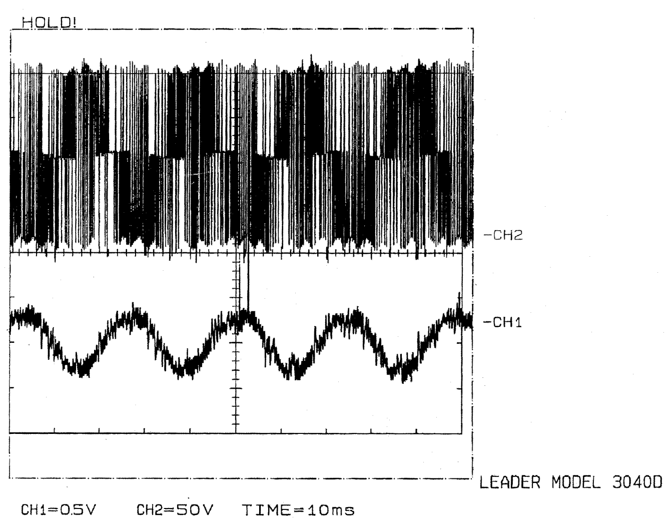

Fig. 11. Typical line voltage 50 V/div (upper) and phase current 5.5 A/div (lower).

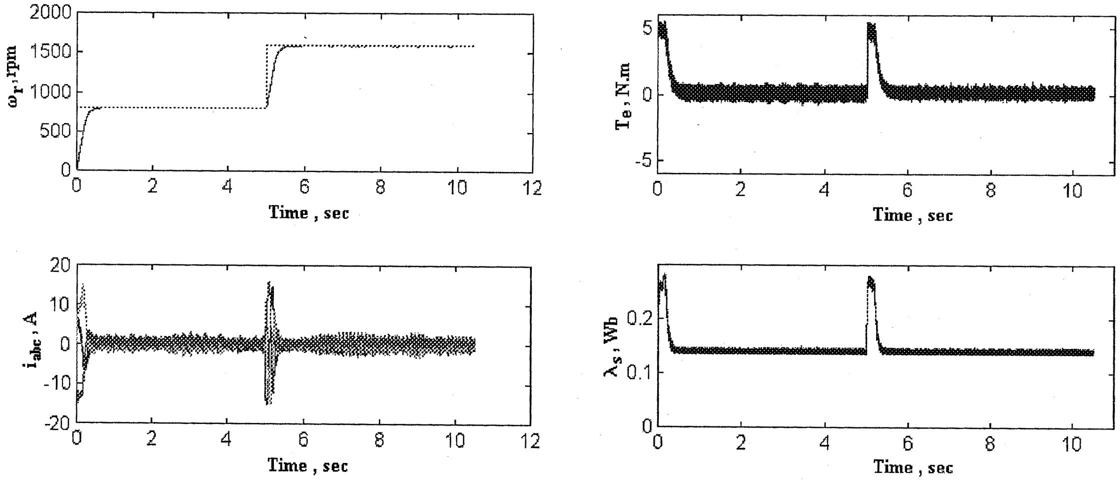

Fig. 12. Simulation results with actual dc supply.

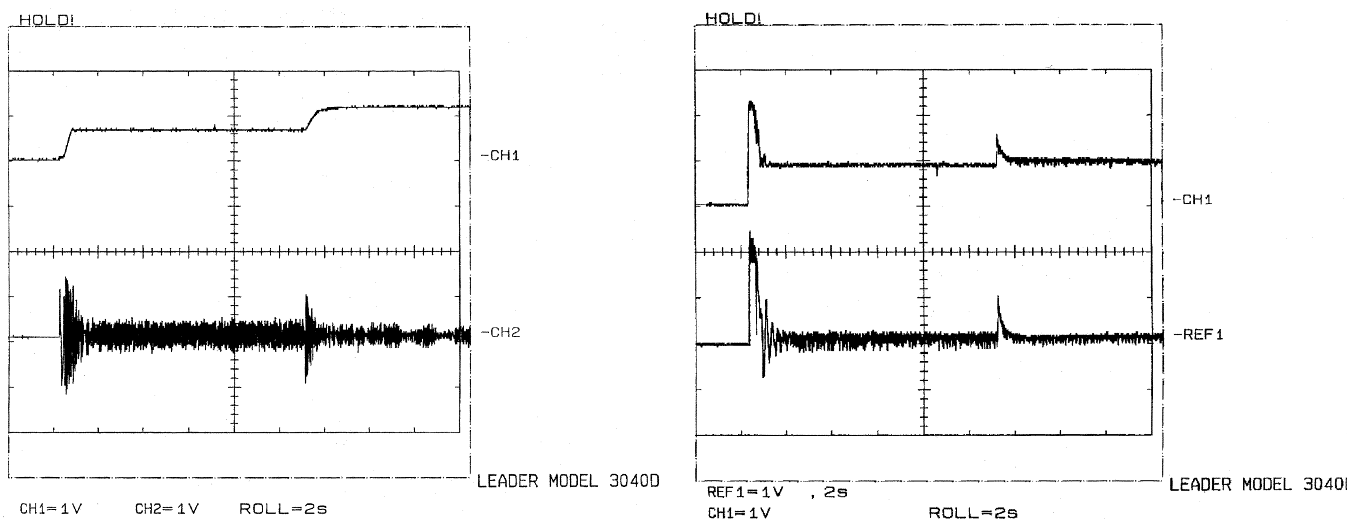

Fig. 13. Experimental results. (a) Upper: speed (CH1:1334 r/min/V); lower: phase current (CH2:11.2 A/V. (b) Upper: flux (CH1:0.156 Wb/V); lower: torque (REF1:2.5 N·m/V).

results. Since T_{cogg} of the proposed PMSM was unknown, the torque pulsation in the simulation results is larger than that of the experimental results. This means that T_{cogg} has been exaggerated in the simulation.

VIII. Conclusions

A new analytical technique has been developed for generating a reference flux from the torque, following the MTPF trajectory. The geometrical mean of the reference estimated torques has been used for reference flux generation, which: 1) removes the constraint $\lambda_n \leq 1$ and 2) since the flux does not only vary with the instantaneous torque, the motor will be stable. The simulation results of the salient-pole PMSM, using the MTPF for obtaining the reference flux, indeed verifies the proposed control technique. In this technique, the flux has been analytically calculated from the torque with a good approximation, thereby eliminating the need for a lookup table. The results of the application of the MTPA and MTPF techniques to the same motor have been compared by means of the simulation process.

Comparison of the experimental and simulated results confirm the proposed method. The tests show that a PMSM drive can be developed using a high-speed processor such as a Pentium 200 MMX instead of a costly floating-point DSP. Clearly, use of a DSP may lead to more precise control. Moreover, monitoring and updating routines for the motor parameters can be implemented simultaneously with the control operation.

References

[1] I. Takahashi and T. Noguchi, "A new quick-response and high-efficiency control strategy of an induction motor," *IEEE Trans. Ind. Applicat.*, vol. IA-22, pp. 820–827, Sept./Oct. 1986.

[2] Y. A. Chapuis, D. Roye, and J. Davoine, "Principles and implementation of direct torque control by stator flux orientation of an induction motor," in *Proc. IEEE PESC'95*, 1995, pp. 185–191.

[3] L. Zhong *et al.*, "Analysis of direct torque control in permanent magnet synchronous motor drives," *IEEE Trans. Power Electron.*, vol. 12, pp. 528–536, May 1997.

[4] M. F. Rahman, L. Zhong, and K. W. Lee, "A direct torque controlled interior permanent magnet synchronous motor drive incorporating field weakening," *IEEE Trans. Ind. Applicat.*, vol. 34, pp. 1246–1253, Nov./Dec. 1998.

[5] C. French and P. Acarnley, "Direct torque control of permanent magnet drive," in *Conf. Rec. IEEE-IAS Annu. Meeting*, vol. 1, Lake Buena Vista, FL, 1995, pp. 199–206.
[6] P. Vas, *Sensorless Vector and Direct Torque Controls*. London, U.K.: Oxford Univ. Press, 1998.
[7] P. Brandstetter and M. Mech, "Direct torque and reactive power control of permanent magnet synchronous motor drive with signal processor," in *Proc. EPE'94*, vol. D1.1-3, Florence, Italy, 1994, pp. 93–96.
[8] M. F. Rahman *et al.*, "A direct torque controlled permanent magnet synchronous motor drive without a speed sensor," in *Proc. IEMD'99*, 1999, pp. 123–125.

Jawad Faiz (M'90–SM'93) received the B.Sc. and M.Sc. degrees with First Class Honors from Tabriz University, Tabriz, Iran, in 1974 and 1975, respectively, and the Ph.D. degree from the University of Newcastle upon Tyne, Newcastle upon Tyne, U.K., in 1988, all in electrical engineering.

Early in his career, he served as a Faculty Member at Tabriz University for ten years. After receiving the Ph.D. degree, he joined Tabriz University, where he held the position of Assistant Professor from 1988 to 1992, Associate Professor from 1992 to 1997, and has been a Professor since 1998. Since February 1999, he has been a Professor in the Department of Electrical and Computer Engineering, Faculty of Engineering, University of Tehran, Tehran, Iran. He is the author of 61 publications in international journals and 68 publications in international conference proceedings. His teaching and research interests are switched reluctance and variable reluctance motor design, and design, modeling, and control of electrical machines, drives, and transformers.

Dr. Faiz is a Member of the IEEE Power Engineering, IEEE Industry Applications, IEEE Power Electronics, IEEE Education, IEEE Industrial Electronics, and IEEE Magnetics Societies. He is also a Member of the Iran Academy of Sciences.

S. Hossein Mohseni-Zonoozi received the B.Sc. degree in electronics engineering from the University of Petroleum Industry, Ahwaz, Iran, in 1996, and the M.Sc. degree in electrical power engineering from the University of Tabriz, Tabriz, Iran, in 1999.

He is currently doing his mandatory military service in Tehran, Iran. His research interests are in electrical machine drives and power electronic applications.